

Asset Analysis: test

Asset ID:	2	Assessment Date:	2026-07-13
Location:	test	Status:	APPROVED & REVIEWED

1. Executive Summary

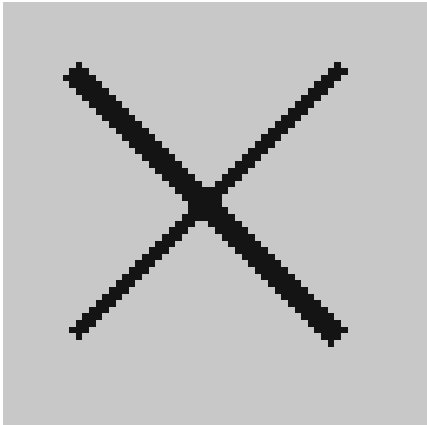
The asset, located at test, has been diagnosed with a medium risk level based on the time-series analysis and a damaged status due to cracks/scratches detected through visual inspection. The overall health status of the asset indicates a need for immediate attention to prevent further deterioration. The risk level is primarily influenced by temperature, vibration, and pressure anomalies, which collectively contribute to the medium risk assessment.

2. Diagnostic Predictions

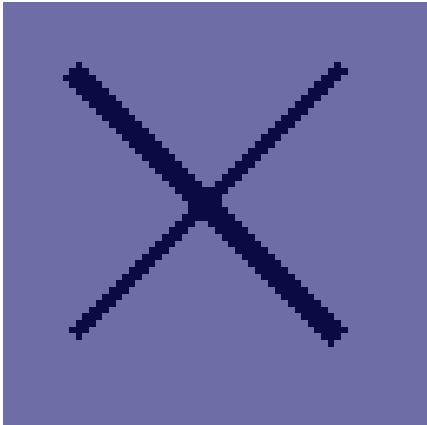
Diagnostic Module	Initial AI Prediction	Final Adjusted Status
Tabular Time-Series (LSTM)	Medium Risk	Medium Risk
Visual Damage (CNN)	Damaged (Cracks/Scratches)	Damaged (Cracks/Scratches)
Sound Analysis (ANN)	Normal Noise	Normal Noise

3. Explainable AI (XAI) Insights

Time-Series Perturbation Saliency: Temperature: 40.6% impact • Vibration: 32.3% impact • Pressure: 27.2% impact



Original Image



Visual Grad-CAM Saliency Highlight

4. Technical Risk Reasoning

The technical reasoning behind the findings suggests that the asset's medium risk level is attributed to the interplay of various factors, including time-series anomalies detected by the LSTM model. The model highlights the importance of temperature (0.4057190716266632), vibration (0.3225538730621338), and pressure (0.271727055311203) in contributing to the risk level, indicating that these factors are critical in assessing the asset's health. Furthermore, the CNN-based visual inspection reveals cracks/scratches on the asset, which, despite a moderate confidence level of 0.50, signifies a potential point of failure. In contrast, the machine sound diagnostic (ANN) indicates a normal machine sound, suggesting that the asset's operational performance is currently unaffected by the detected anomalies.

5. Recommended Mitigation & Compliance Steps

- To mitigate the identified risks and ensure compliance with regulatory guidelines, the following action steps are proposed:
- Conduct a thorough maintenance check to address the detected cracks/scratches, focusing on repairing or replacing damaged components to prevent further deterioration.
- Implement a real-time monitoring system to track temperature, vibration, and pressure levels, enabling prompt identification of anomalies and allowing for proactive measures to prevent asset failure.
- Develop and schedule regular inspection routines, incorporating both visual inspections using CNN-based tools and audio diagnostics, to ensure the asset's condition is continuously assessed and maintained in accordance with regulatory standards, thereby minimizing the risk of unexpected failures and ensuring compliance with safety and operational regulations.

6. Human-in-the-Loop Analyst Verification

Analyst Remarks: No human analyst notes provided.

Reviewed By: sharjeel ahmed

Review Date: 2026-07-13